

E1

Consider the circuit in Fig. 1. The circuit parameters at $T = 25^\circ\text{C}$ are the following:

$V_{DD} = 10\text{ V}$; $R_{G1} = 150\text{ k}\Omega$; $R_{G2} = 100\text{ k}\Omega$; $R_{G3} = 100\text{ k}\Omega$; $R_{S1} = 1\text{ k}\Omega$; $R_{D2} = 15\text{ k}\Omega$; $R_F = 15\text{ k}\Omega$; $R_S = 1\text{ k}\Omega$; $C_S = 10\text{ }\mu\text{F}$; $C_G = 10\text{ }\mu\text{F}$; $C_{S1} = 10\text{ }\mu\text{F}$.

M_1 (E-NMOS): $V_{t1} = 2\text{ V}$; $I_{DSS1} = 2.4\text{ mA}$; $r_0 = +\infty$;

M_2 (E-PMOS): $V_{t2} = -3\text{ V}$; $I_{DSS2} = 3.6\text{ mA}$; $r_0 = +\infty$;

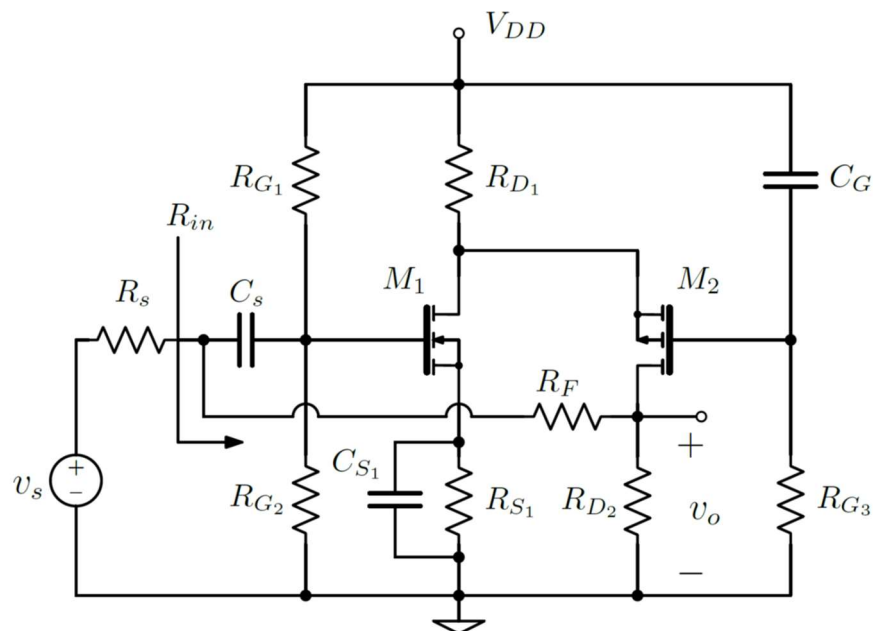


Fig. 1 – A case of two stage amplifier.

Considering all capacitors to be equivalent to *open circuits*, determine:

1. the value of R_{D1} such that the voltage at the drain of M_2 is $V_O = 2\text{ V}$ in the steady state;
2. the operating points (V_{DS} , I_{DS}) of M_1 and M_2 .

Assuming C_S , C_{S1} and C_G to be equivalent to *short circuits* at the frequencies of interest (i.e. at mid-band), determine also:

3. the voltage gain $A_v = v_o/v_s$ of the amplifier;
4. the input resistance indicated in the figure.

Applying the short circuit time constant method, determine, finally:

5. the low frequency bandwidth limit f_L of the amplifier.

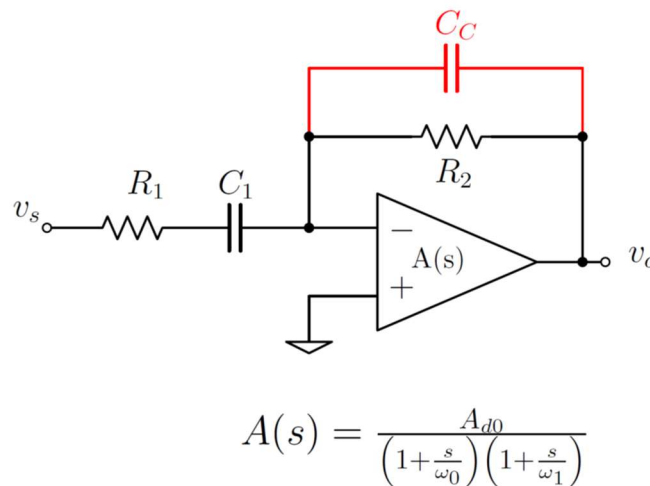


Fig. 2 – Partially compensated OpAmp with high pass frequency response.

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R_1 = 10 \text{ k}\Omega$; $R_2 = 220 \text{ k}\Omega$; $C_1 = 100 \text{ nF}$;

OpAmp: $A_{d0} = 10^5 \text{ [V/V]}$; $\omega_0 = 10 \text{ rad/s}$; $\omega_1 = 5000 \text{ rad/s}$.

Considering capacitor C_C equivalent to an open circuit, determine:

1. the expression of the *ideal* circuit voltage gain $A_v(s) = v_o/v_s$;
2. an estimation of the circuit *phase margin*;
3. the complete *block diagram* of the amplifier, according to *feedback theory*;
4. the expression of the *input scaling function* $k_s(s)$ in the above diagram.

Then, define

5. the maximum value of capacitor C_C for which the amplifier maintains *approximately* the *same frequency response* of the original circuit *up to the crossover frequency*;
6. the *new phase margin* of the circuit, considering capacitor C_C equal to the value found at point 5. (hint: the compensated beta network shows a low frequency pole and a high frequency pole, the latter falling well above the crossover frequency)